PROJECT REPORT

1. INTRODUCTION

1.1 Project Overview

ToyCraft Tales – *Tableau's Vision into Toy Manufacturer Data* is a data visualization and business intelligence project developed using **Tableau**. The project focuses on analyzing toy manufacturing and sales data to generate meaningful insights that support business decision-making.

In the toy manufacturing industry, large amounts of data are generated daily, including sales records, product categories, customer demand, regional distribution, profit margins, and inventory details. Managing and analyzing this raw data manually can be time-consuming and complex.

This project transforms raw toy manufacturer data into interactive dashboards using Tableau. These dashboards include visual elements such as bar charts, line graphs, pie charts, KPI indicators, and region-wise maps. By using these visual tools, stakeholders can easily understand business performance, identify trends, and make informed strategic decisions.

ToyCraft Tales aims to bridge the gap between raw data and actionable insights through effective data visualization techniques.

1.2 Purpose

The primary purpose of this project is to analyze and visualize toy manufacturer data in a structured and meaningful way. The project is designed to:

- Understand overall sales performance.
- Identify best-selling toy categories.
- Analyze profit and loss trends.
- Compare region-wise sales distribution.
- Study seasonal sales patterns.
- Support data-driven business decisions.

By implementing this solution using Tableau, the project helps toy manufacturers improve operational efficiency, optimize product strategies, and increase profitability.

The ultimate goal of ToyCraft Tales is to demonstrate how data visualization can convert complex datasets into clear, interactive, and insightful dashboards that enhance business intelligence.

2. IDEATION PHASE

2.1 Problem Statement

Toy manufacturing companies generate large amounts of production, inventory, sales, and quality data, which is often stored in raw formats like Excel files or databases. This makes analysis difficult and time-consuming for business users. The project focuses on collecting manufacturing data, storing it in an SQL database, and performing data preprocessing using SQL queries. The processed data is then connected to Tableau to create dashboards and story visualizations. These visualizations help users understand trends, performance, and insights easily. The system simplifies data analysis and supports better business decision-making.

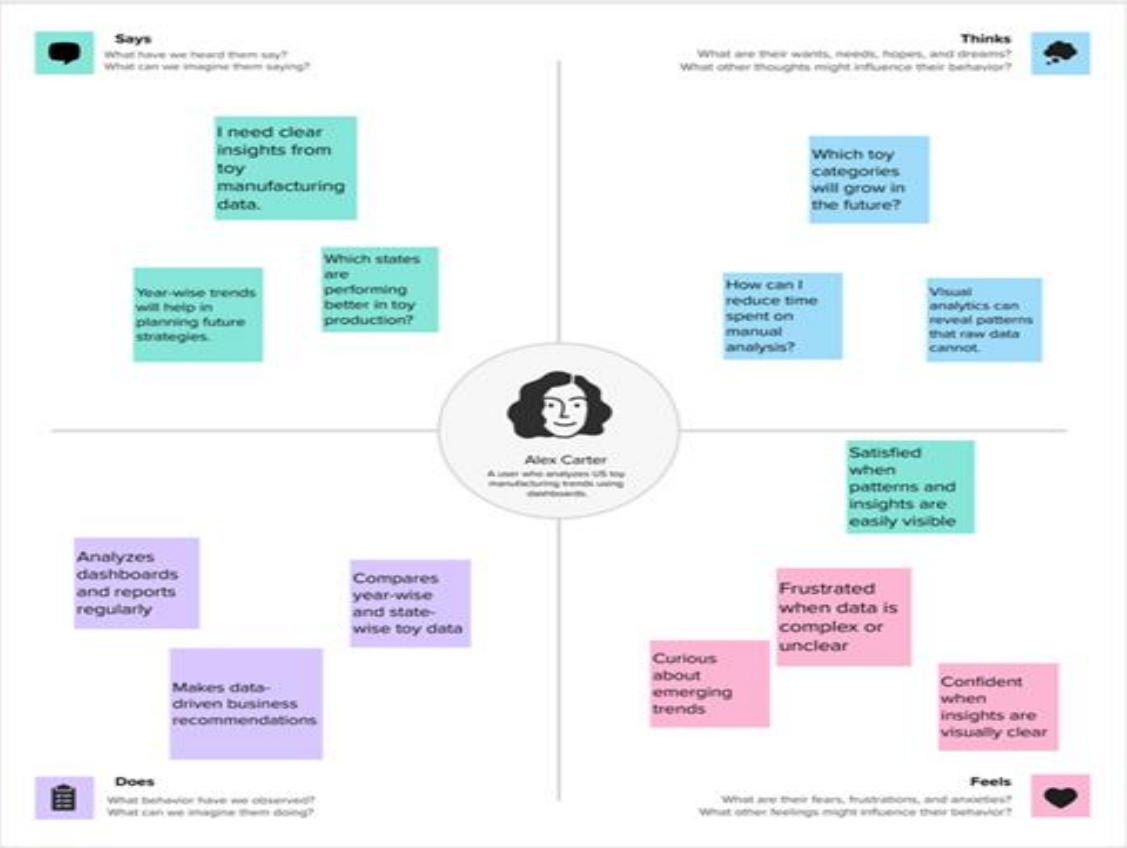
I am	Describe customer with 3-4 key characteristics - who are they?	Describe the customer and their attributes here
I'm trying to	List their outcome or "job" the care about - what are they trying to achieve?	List the thing they are trying to achieve here
but	Describe what problems or barriers stand in the way – what bothers them most?	Describe the problems or barriers that get in the way here
because	Enter the "root cause" of why the problem or barrier exists – what needs to be solved?	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions from the customer's point of view – how does it impact them emotionally?	Describe the emotions the result from experiencing the problems or barriers

Example:

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Business user	analyze toy manufacturing performance data	the data is stored in multiple raw files	there is no centralized database or visualization system	confused and time-consuming to understand reports
PS-2	Data analyst	create dashboards for manufacturing insights	preparing data takes too much manual effort	data cleaning and transformation are not automated	frustrated and inefficient

2.2 Empathy Map Canvas

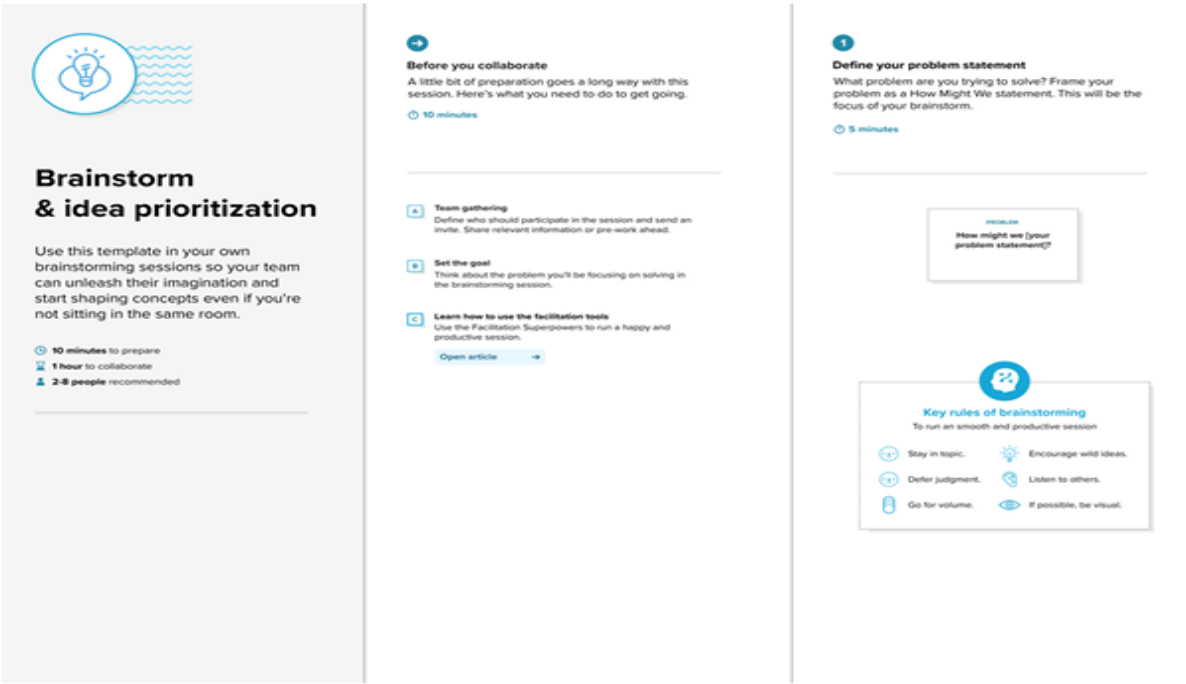
The Empathy Map helps to understand the needs, thoughts, and challenges of the Compare sales reports monthly.



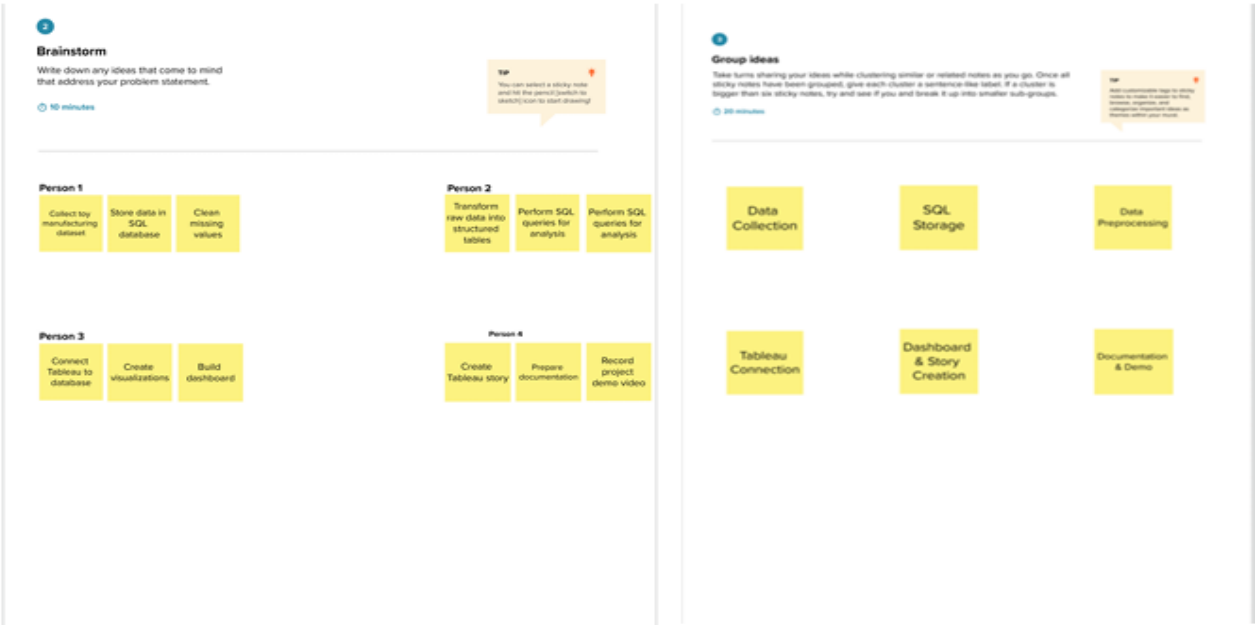
2.3 Brainstorming

Brainstorming helped the team identify problems in analyzing toy manufacturing data stored in different formats. The team discussed ideas for data collection, SQL storage, preprocessing, and visualization. Different solutions were evaluated based on feasibility and usefulness. Important ideas were grouped and prioritized for implementation. This process helped the team plan dashboard and story development in Tableau. Finally, selected ideas were used to build a structured data visualization solution.

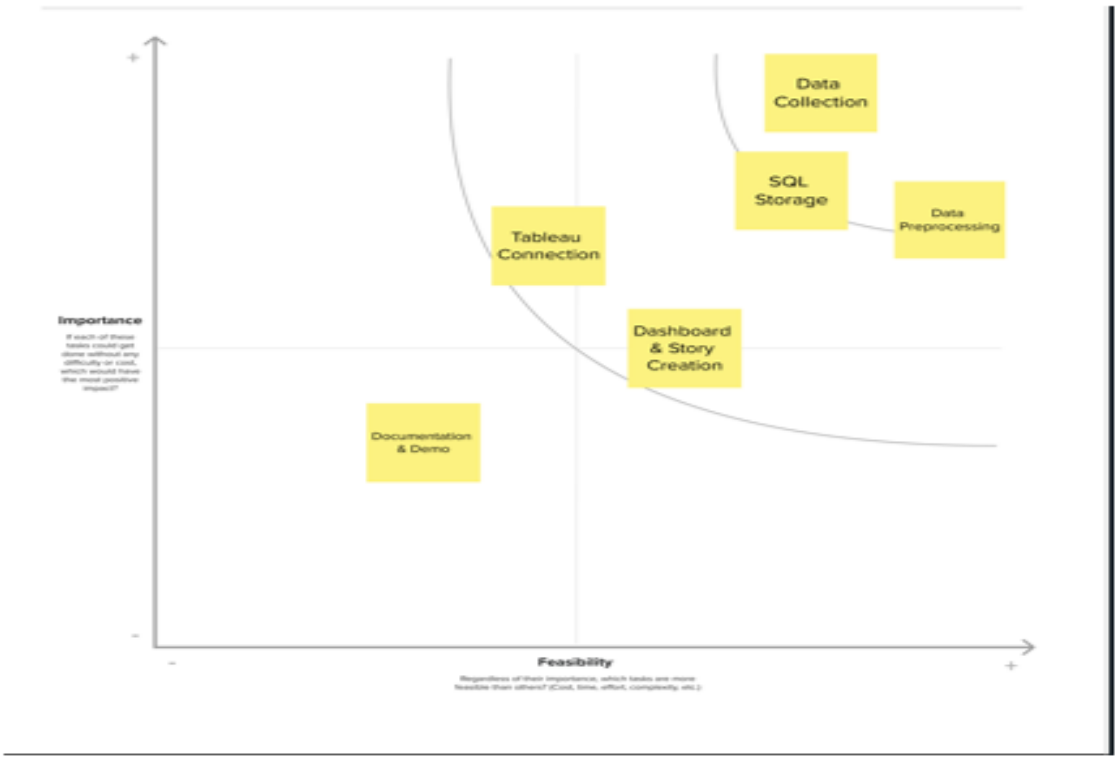
Step-1: Team Gathering, Collaboration and Select the Problem Statement



Step-2: Brainstorm, Idea Listing and Grouping



Step-3: Idea Prioritization



3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The Customer Journey Map explains the step-by-step interaction of the user (Toy Manufacturing Business Manager) with the proposed dashboard system.

Scenario: A user exploring a Tableau dashboard to understand toy manufacturing trends in the US.	Entice How does someone become aware of this service?		Enter What do people experience as they begin the process?	Engage In the core moments in the process, what happens?			Exit What do people typically experience as the process finishes?	Extend What happens after the experience is over?	
Experience steps What does the person (or people) at the center of this scenario typically experience in each step?	Notices the project during internship review or demo.	Discovers the toy manufacturing dashboard while searching for market insights.	Opens the Tableau dashboard.	Analyzes trends by year and region.	Compares state-wise and manufacturer-wise performance	Explores interactive dashboards in detail	Completes data analysis.	Revisits dashboard when needed.	Refers to insights for future analysis
Interactions What interactions do they have at each step along the way? • **People:** Who do they see or talk to? • **Places:** Where are they? • **Things:** What digital touchpoints or physical objects do they use?	Reads project title and brief description.	Reads the dashboard title and overview.	Views charts, graphs, and filters.	Uses filters and hover options.	Switches between charts and views	Interacts with dashboards using mouse and dropdowns	Scrolls through final dashboards.	Shares project with others.	Opens saved dashboards again
Goals & motivations At each step, what is a person's primary goal or motivation? ("Help me..." or "Help me avoid...")	Wants to know what insights the project offers.	Understand whether the dashboard is useful for analysis.	Understand toy manufacturing data.	Compare performance and identify trends.	Get insights to support data-driven decisions	Understand performance differences clearly	Gain overall understanding.	Use insights for reference.	Support future planning and reporting
Positive moments What steps does a typical person find enjoyable, productive, fun, motivating, delightful, or exciting?	Clear project title and purpose.	Clear purpose and clean dashboard layout.	Attractive and interactive visuals.	Clear insights from visuals.	Interactive filters make analysis easy	Patterns are easy to understand visually	Feels informed and confident.	Easy access to information.	Useful for presentations and reviews
Negative moments What steps does a typical person find frustrating, confusing, angering, costly, or time-consuming?	Unclear how the dashboard works.	Limited explanation about the data source.	Initial confusion with charts.	Too much data on one screen.	Some charts may need explanation	Takes time to understand complex views	No summary section.	Cannot export data.	Limited export or sharing options
Areas of opportunity How might we make each step better? What ideas do we have? What have others suggested?	Add a simple introduction and overview.	Provide a short teaser or preview of key insights to attract users.	Add labels and tooltips.	Highlight key findings clearly.	Add short explanations near charts	Simplify complex dashboards	Add a conclusion or summary.	Add download or export options.	Enable dashboard updates with new data

Activate Windows

3.2 Solution Requirement

A. Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Visualization	Import toy manufacturing data into Tableau Create year-wise and state-wise visualizations
FR-2	Interactive Dashboard	Provide filters for year and region Enable interactive charts and tooltips
FR-3	Dashboard Analysis	Analyze trends and patterns Compare toy performance across regions
FR-4	Storytelling	Create story points from dashboards Present insights in a logical sequence
FR-5	Data Filtering and Comparison	Allow users to filter data by year and state Enable comparison between different regions and time periods
FR-6	Insight Generation	Highlight key insights from dashboards Support decision-making through visual analysis

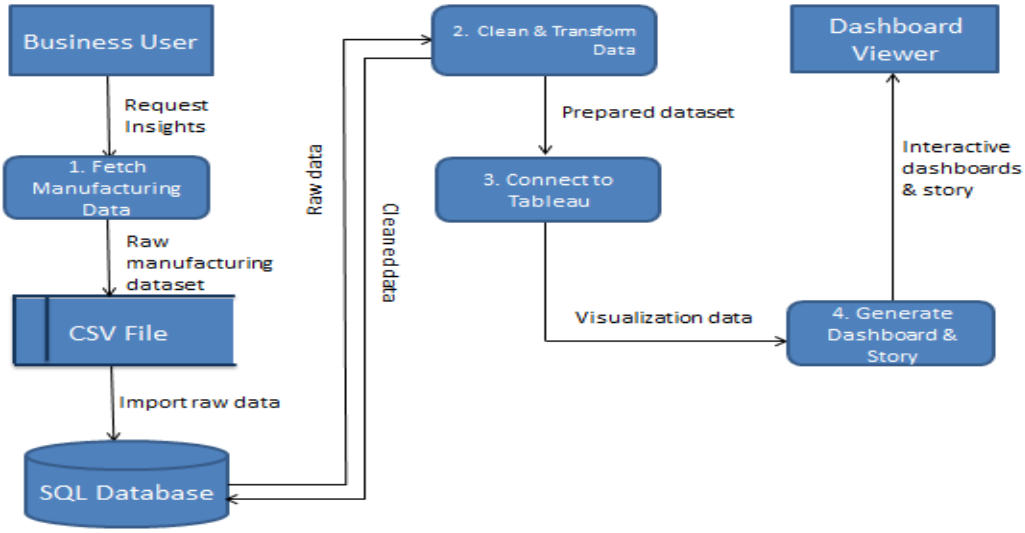
B. Non-Functional Requirements

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The dashboard should be easy to understand and navigate for users.
NFR-2	**Security**	The data used in the dashboard should be handled securely.
NFR-3	**Reliability**	The dashboard should display accurate and consistent results.
NFR-4	**Performance**	The dashboard should load and respond quickly

		to user interactions.
NFR-5	**Availability**	The dashboard should be accessible whenever required.
NFR-6	**Scalability**	The solution should support additional data in the future.

3.3 Data Flow Diagram



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Business User	View Dashboard	USN-1	As a business user, I can view toy manufacturing dashboard	Dashboard loads successfully with charts	High	Sprint-1
Business User	Filter Data	USN-2	As a business user, I can filter data by year and category	Filters update data correctly	High	Sprint-1
Business User	Analyze Trends	USN-3	As a business user, I can analyze sales trends	Trend graphs are visible	Medium	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Business User	View Insights	USN-4	As a business user, I can view insights about production and revenue	Insights are clearly displayed	High	Sprint-1
Business User	Export Reports	USN-5	As a business user, I want to export dashboard reports	Reports download successfully in PDF or image format	Medium	Sprint-2
Business User	Search Data	USN-6	As a business user, I want to search specific product data	Search results display accurate information	Medium	Sprint-2
Data Analyst	Data Cleaning	USN-7	As a data analyst, I can clean raw manufacturing data	Cleaned dataset is stored in database	High	Sprint-1
Data Analyst	Data Transformation	USN-8	As a data analyst, I can transform data for visualization	Processed dataset is ready for Tableau	High	Sprint-1
Administrator	Database Management	USN-9	As an admin, I can manage SQL database	Database updates correctly	Medium	Sprint-2
Administrator	Monitor System	USN-10	As an admin, I can monitor dashboard performance	System runs without errors	Medium	Sprint-2
Administrator	Data Backup	USN-11	As an admin, I want to backup the database regularly	Backup is created successfully without data loss	Medium	Sprint-2
Administrator	Access Control	USN-12	As an admin, I want to control dashboard access permissions	Only authorized users can access the dashboard	High	Sprint-2

3.4 Technology Stack

Technical Architecture:

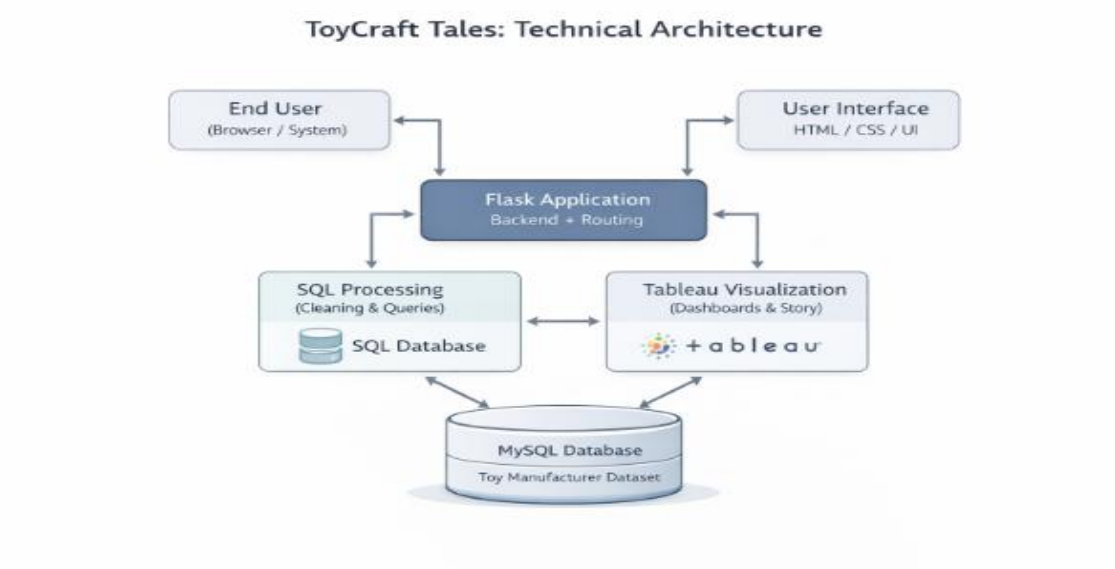


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application e.g. Web UI, Mobile App, Chatbot etc.	HTML, CSS, Tableau
2.	Application Logic-1	Logic for a process in the application	Python
3.	Application Logic-2	Logic for a process in the application	SQL
4.	Application Logic-3	Logic for a process in the application	_
5.	Database	Data Type, Configurations etc.	MySQL
6.	Cloud Database	Database Service on Cloud	_
7.	File Storage	File storage requirements	Local File System
8.	External API-1	Purpose of External API used in the application	_

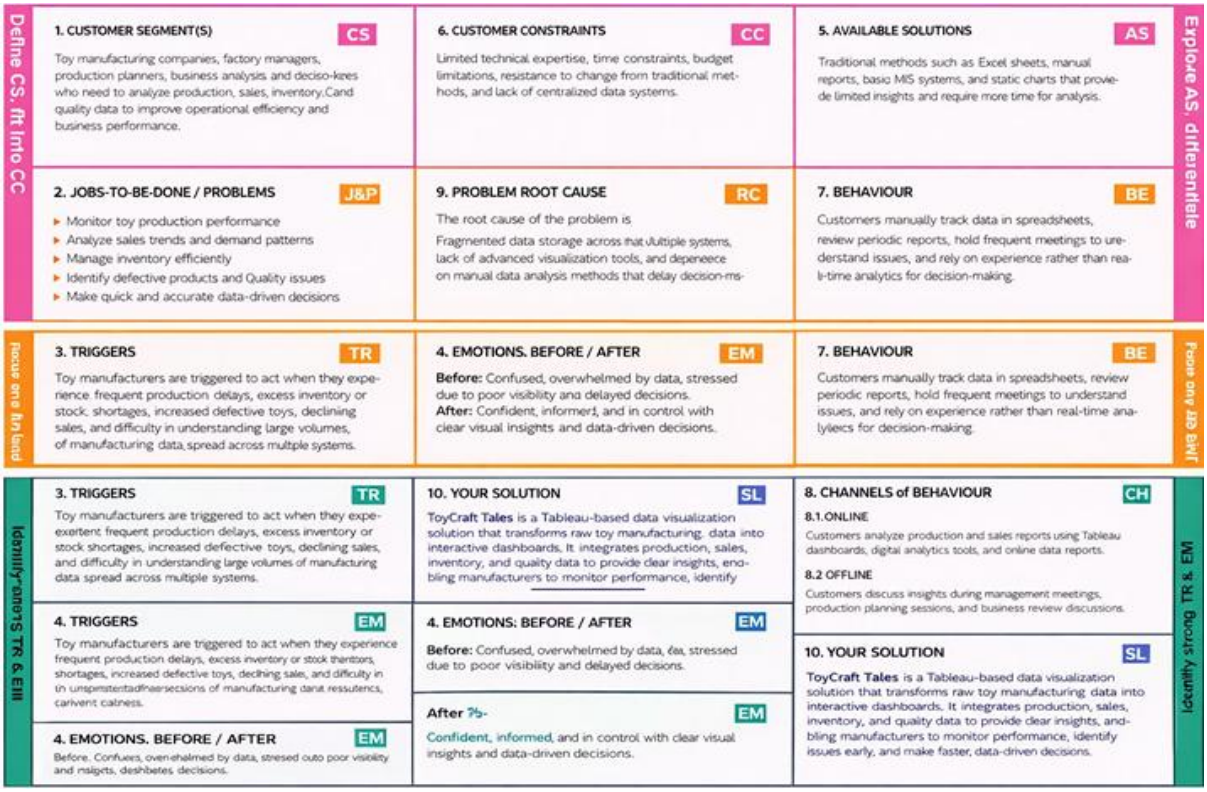
9.	External API-2	Purpose of External API used in the application	_
10.	Machine Learning Model	Purpose of Machine Learning Model	Tableau Public 2025.3
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration :	Local Server.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	Python, Flask, VS Code , Tableau Desktop Public Edition
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	_
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	_
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	Locally Installed Tools (Flask, MySQL, Tableau Desktop)
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	Optimized SQL Queries, Indexed Database, Tableau Data Filtering

4. PROJECT DESIGN

4.1 Problem Solution Fit



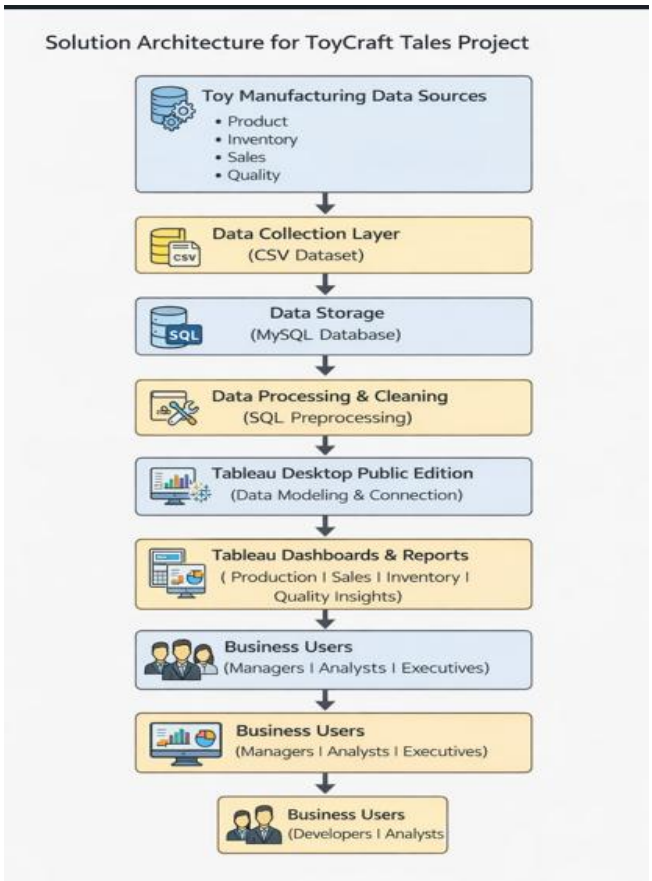
4.2 Proposed Solution

S.No	Parameter	Description
1	**Problem Statement**	The toy manufacturing industry produces large volumes of production, sales, inventory, and quality data. Due to lack of proper visualization and analytics tools, manufacturers find it difficult to analyze data efficiently, leading to poor decision-making, higher costs, and operational inefficiencies.
2	**Idea / Solution Description**	*ToyCraft Tales* is a Tableau-based data visualization solution that transforms raw toy manufacturing data into interactive dashboards. It enables manufacturers to monitor production performance, sales trends, inventory levels, and quality metrics in a single platform.
3	**Novelty / Uniqueness**	The solution integrates multiple data sources into Tableau dashboards, provides real-time interactive visualizations, supports drill-down analysis, and presents data through visual storytelling that is easy for non-technical users to understand.
4	**Social Impact / Customer Satisfaction**	Improves toy quality by identifying defects early, reduces waste and overproduction, ensures timely product availability, and enhances customer satisfaction through better demand forecasting.

5	**Business Model (Revenue Model)**	Revenue is generated through subscription-based dashboard access, customized Tableau dashboard development, enterprise licensing, and data analytics consultancy services.
6	**Scalability of the Solution**	The solution is highly scalable and can handle large datasets, support multiple manufacturing units, integrate with cloud data sources, and be extended to other manufacturing domains beyond the toy industry.

4.3 Solution Architecture

The solution architecture for ToyCraft Tales: Tableau's Vision into Toy Manufacturing Data bridges the gap between toy manufacturing business problems and data-driven technology solutions. It identifies the best use of Tableau to analyze and visualize production, sales, inventory, and quality data. The architecture defines the system structure, data flow, and key features, and provides clear specifications for developing, managing, and delivering an efficient and scalable analytics solution.



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Product Backlog, Sprint Schedule, and Estimation

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Collecting Data	USN-1	As an analyst, I gather toy manufacturing dataset from available sources.	3	High	Tirusula Kalanjali
Sprint-1	Data Preprocessing	USN-2	As an analyst, I prepare and organize raw data for analysis.	2	High	Y G D Manasa
Sprint-2	Build Visualization	USN-3	As an analyst, I create charts and visualizations in Tableau.	3	High	Y G D Manasa
Sprint-2	Dashboard	USN-4	As a user, I design an interactive dashboard to display insights.	2	Medium	Y G D Manasa
Sprint-3	Story	USN-5	As a user, I build a Tableau story to present findings clearly.	2	High	Y G D Manasa
Sprint-3	Web Integration	USN-6	As a developer, I integrate Tableau output with the project interface.	3	High	Y G D Manasa
Sprint-4	Work on Templates	USN-7	As a team member, I prepare project templates and final formatting.	2	Medium	Team
Sprint-4	Final Documentation & Video	USN-8	As a team member, I complete documentation and record project video.	2	High	Pasalapudi S S D M Akshitha

Project Tracker, Velocity & Burndown Chart:

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	5	2 Days	4 Feb 2026	05 Feb 2026	5	05 Feb 2026

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-2	5	3 Days	06 Feb 2026	08 Feb 2026	5	08 Feb 2026
Sprint-3	5	3 Days	09 Feb 2026	11 Feb 2026	5	11 Feb 2026
Sprint-4	4	2 Days	12 Feb 2026	13 Feb 2026	4	13 Feb 2026

Velocity:

Sprint Duration = 10 days

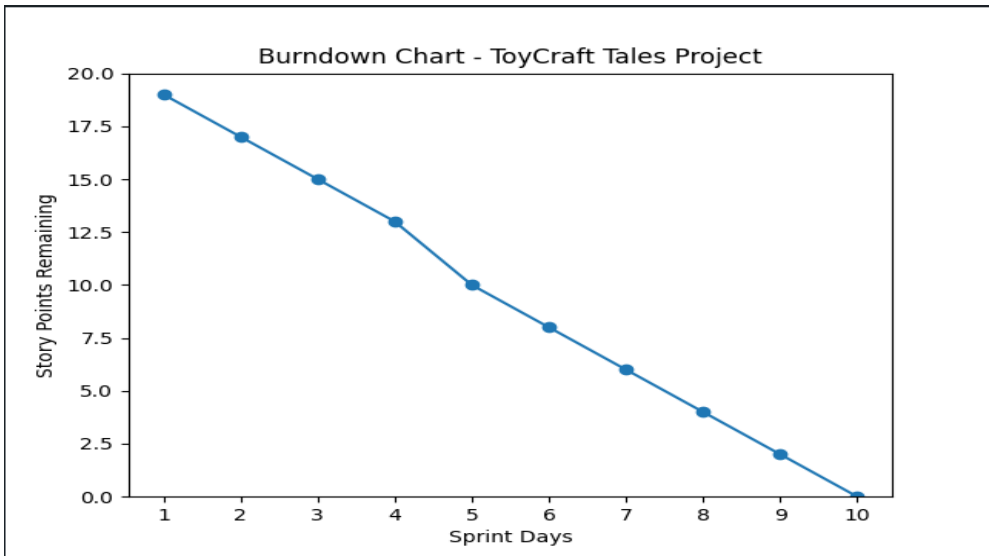
Total Story Points = 19

Velocity = 19 / 10

Velocity = 1.9 story points per day

Burndown Chart:

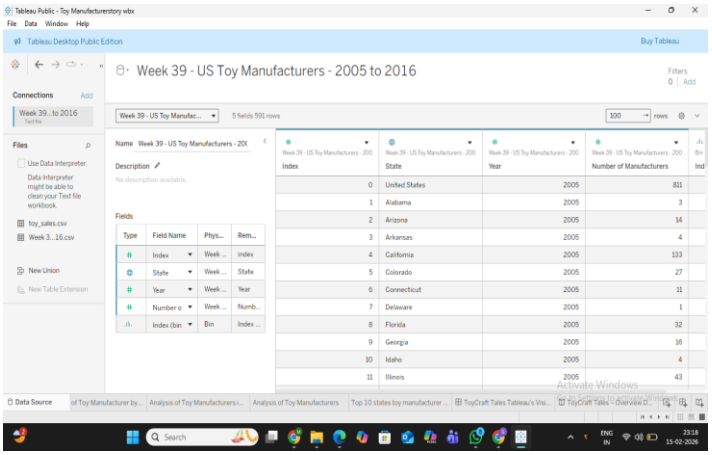
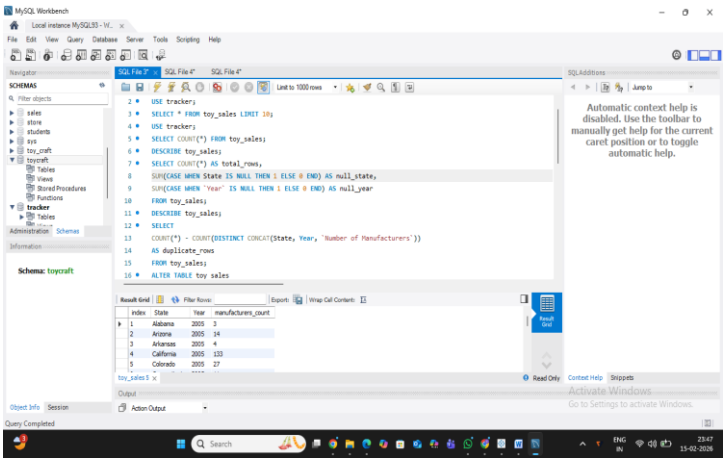
The burndown chart shows the progress of the ToyCraft Tales project across sprint days.
It represents the remaining story points decreasing as tasks are completed.
The steady downward trend indicates that the team completed work consistently during each sprint.
This chart helps track project progress and ensures timely completion of tasks.

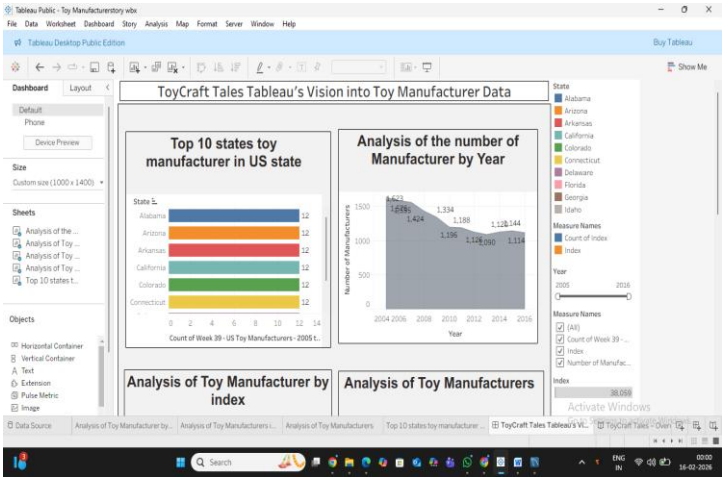
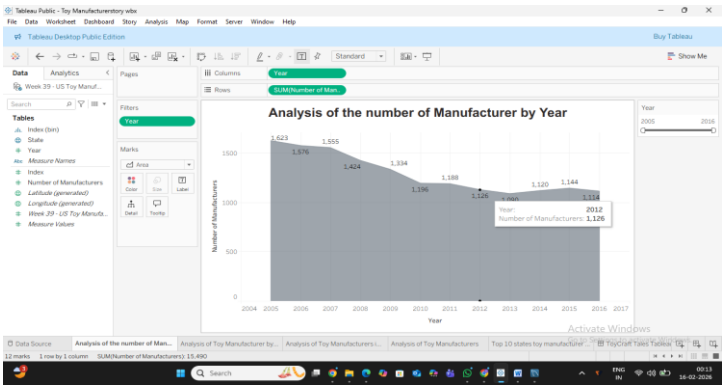


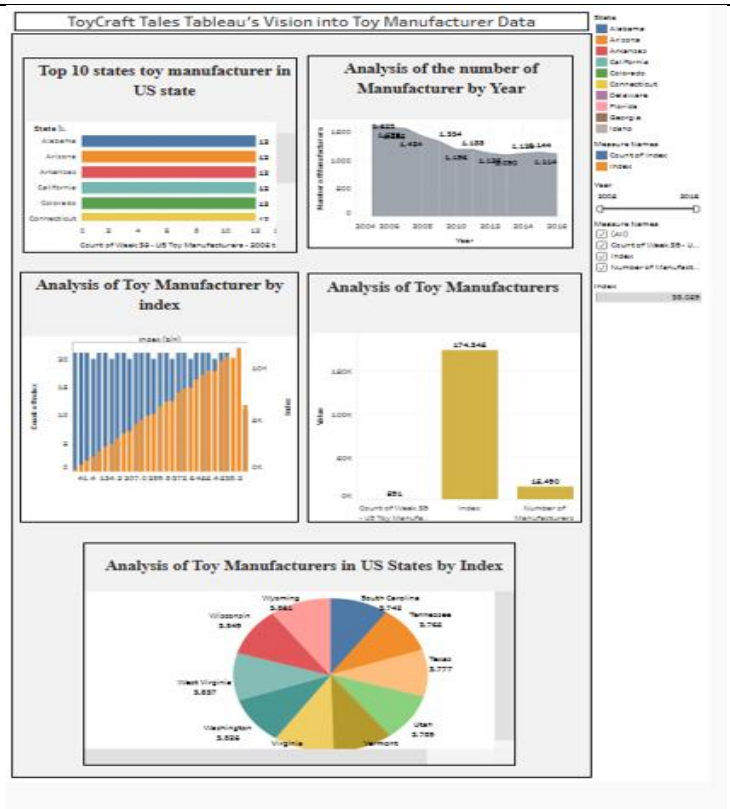
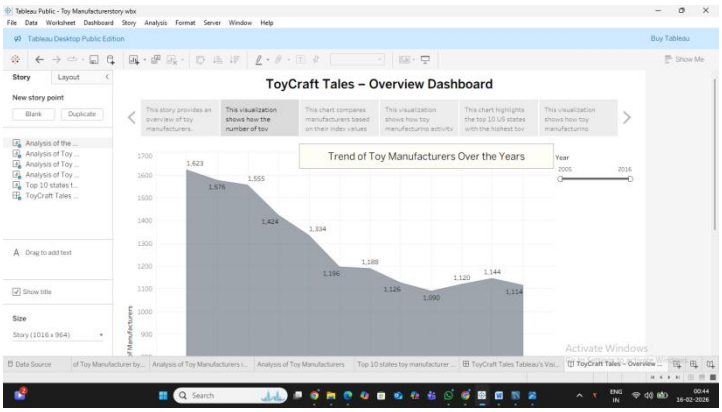
6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Model Performance Testing:

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	The dataset “Week 39 – US Toy Manufacturers (2005–2016)” was successfully loaded into Tableau. It contains **591 rows and 4 fields** (State, Index, Number of Manufacturers, and Year). The data shows the number of toy manufacturers in different US states from 2005 to 2016. 
2.	Data Preprocessing	Data cleaning and validation were performed using SQL queries in MySQL Workbench. NULL values and duplicate records were checked, and the table structure was verified before connecting the dataset to Tableau. 

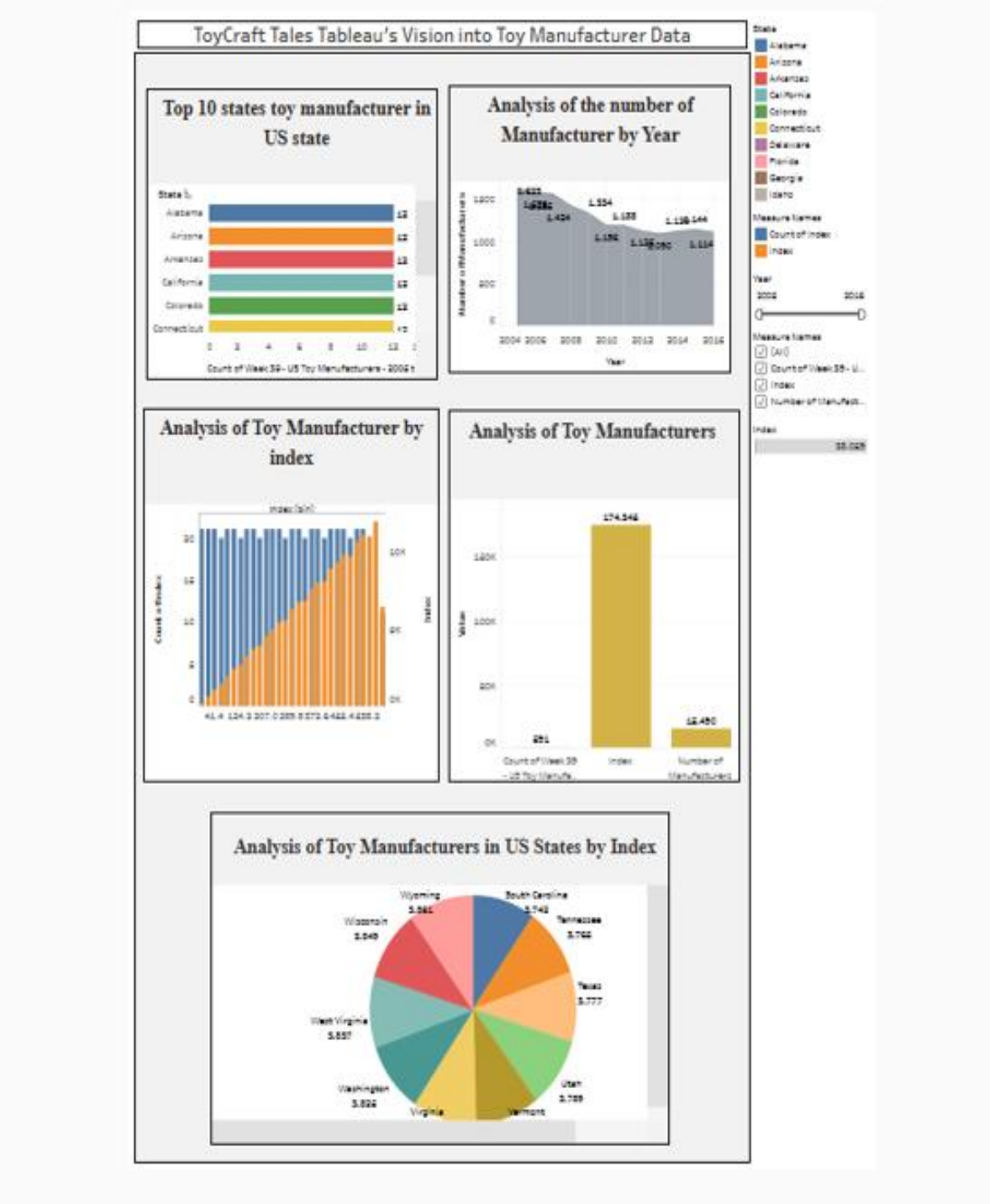
3.	Utilization of Filters	Filters such as State and Year were applied in the dashboard to enable interactive data analysis. Users can dynamically select specific values to update the visualizations. 
4.	Calculation fields Used	**No of Calculation Fields: 0** No additional calculated fields were created in the project. Built-in measures such as SUM(Number of Manufacturers) were used for analysis and visualization. 
5.	Dashboard design	No of Visualizations / Graphs :5 The dashboard contains 5 visualizations including bar charts, area chart, and pie chart. The layout is designed clearly with interactive filters for better analysis.

		
6	Story Design	The story contains 6 story points, including one complete dashboard view and individual analysis views for each visualization. 

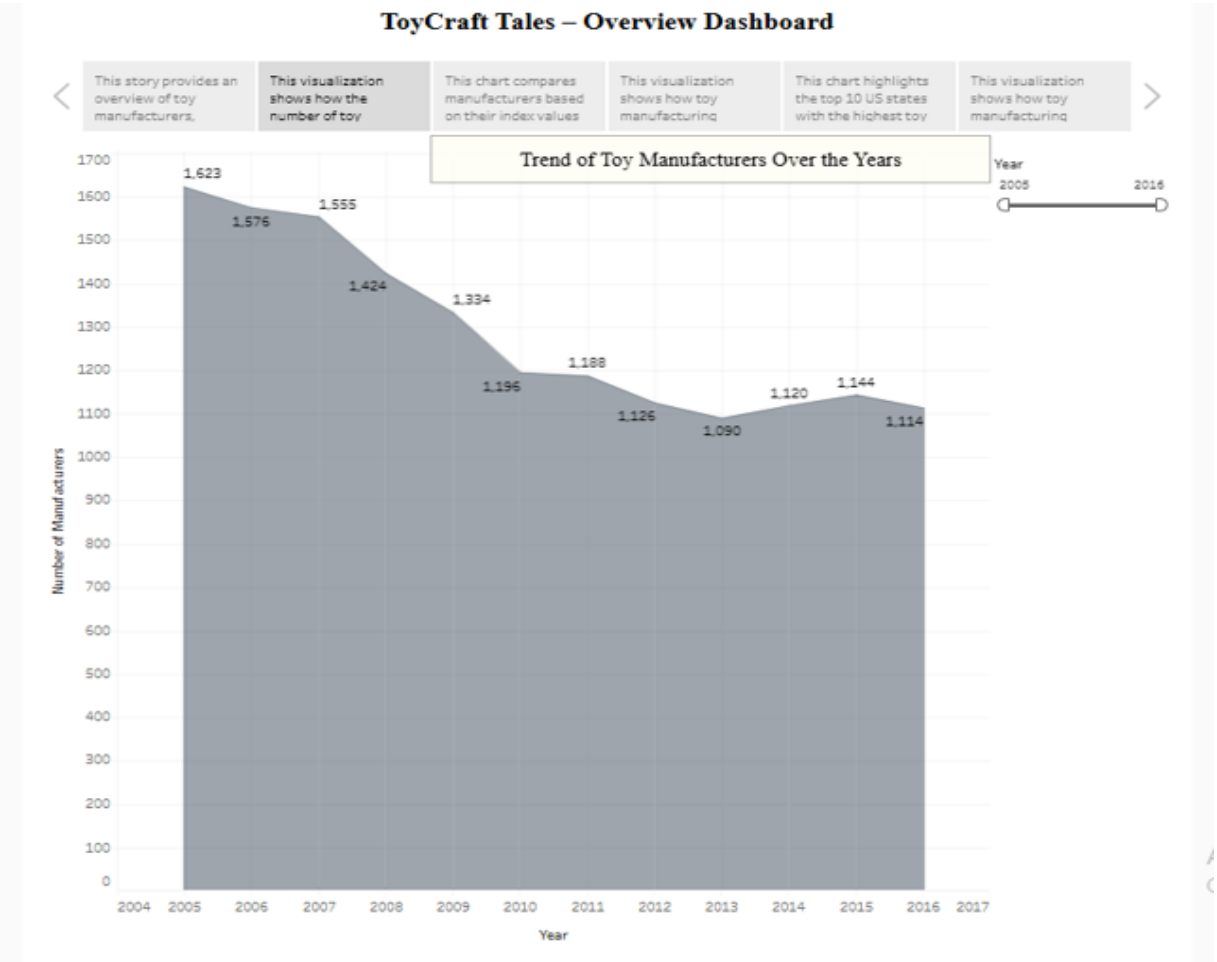
7. RESULTS

7.1 Output Screenshots

Dashboard:



Story:



8. ADVANTAGES & DISADVANTAGES

Advantages

- Converts raw data into clear visual insights.
- Interactive dashboard for easy analysis.
- Saves time compared to manual reporting.
- Helps in better decision-making.
- Improves business performance tracking.

Disadvantages

- Requires licensed software like **Tableau**.
- Performance may reduce with very large datasets.
- Requires basic knowledge of data visualization tools.

9. CONCLUSION

ToyCraft Tales successfully demonstrates how data visualization can improve decision-making in the toy manufacturing industry.

By using Tableau dashboards, raw production and sales data were transformed into meaningful insights. The project helps identify trends, improve performance monitoring, and support data-driven business strategies.

10. FUTURE SCOPE

- Integration with live databases for real-time updates.
- Implementation of predictive analytics for sales forecasting.
- Development of mobile-friendly dashboards.
- Expansion to other manufacturing industries.
- Integration with ERP or cloud-based systems.

11. APPENDIX

Source Code(if any)

Dashboard:

https://public.tableau.com/views/ToyManufacturerwbx/ToyCraftTalesTableausVisioninto ToyManufacturerData?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Story:

https://public.tableau.com/views/ToyManufacturerstorywbx/ToyCraftTalesOverviewDashboard?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Dataset Link

Toy Manufacturers in US States

GitHub & Project Demo Link

GitHub Link:

Project Demo Link:

https://drive.google.com/file/d/1bwWeGYrq0g1A_g76BYsKMO_JbwOMEHxN/view?usp=sharing